

Week 10: Advanced Avellaneda-Stoikov, Structured Products & Finals

Extensions, the corporate view, and communicating results

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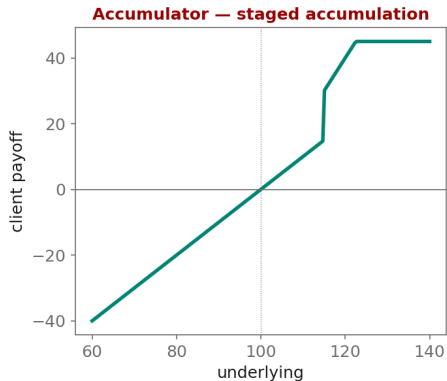
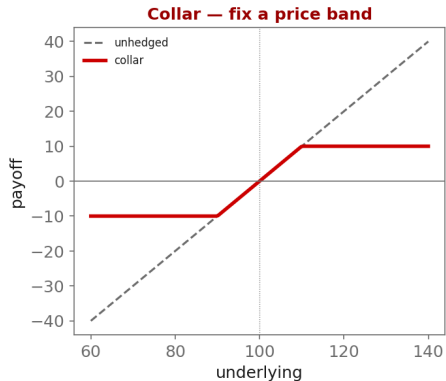
1. Learning objectives
2. Lecture 1: Advanced Extensions
3. Lecture 2 (Guest: Fenni Kang, Coincall): Structured Products
4. Illustration
5. Derivation & rationale
6. Getting the data from Coincall
7. Code: `week10_collar.py`
8. Glossary of key terms
9. Reading
10. References

- ▶ Implement one advanced Avellaneda-Stoikov extension
- ▶ Understand corporate structured products
- ▶ Write and present a research-grade report

- ▶ Multi-asset market making (BTC + ETH, correlated)
- ▶ Adverse-selection-aware and signal-driven quoting
- ▶ Robust optimization: worst-case inventory penalty

- ▶ Collars to fix a price band
- ▶ Accumulators / decumulators for staged flow
- ▶ Coupon / yield-enhancement products
- ▶ How that flow lands on the market-maker's book

Corporate Structured Products (Coincall guest)



$$\min_{\delta} \max_{\Sigma \in \mathcal{U}} \left\{ \text{spread P\&L} - \frac{1}{2} q^T \Sigma q \right\}$$

Why: Robust market making: we do not trust a single covariance estimate Σ , so we optimize against the WORST case over an uncertainty set $\mathcal{U}$. This widens quotes exactly where parameter risk is highest – a principled cure for over-fitting the penalty matrix.

- ▶ Multi-asset extension needs BTC and ETH chains together:
 - ▶ `GET /open/option/get/v1/BTC` and `GET /open/option/get/v1/ETH`
- ▶ Correlation from perp klines (BTC vs ETH) over your backtest window
- ▶ Reuse all Milestone-3 Parquet datasets; no new sources required

Cross-check exact paths at <https://docs.coincall.com/>


```
import numpy as np
def collar_payoff(S, spot=100.0, put_K=90.0, call_K=110.0):
    """Long underlying + long put(put_K) + short call(call_K): a fixed band."""
    return (S - spot) + np.maximum(put_K - S, 0.0) - np.maximum(S - call_K, 0.0)

S = np.linspace(60, 140, 9)
print(dict(zip(S.round(0), collar_payoff(S).round(1)))) # capped above, floored below
```

Multi-asset market making jointly quoting correlated underlyings (e.g., BTC and ETH)

Robust optimization optimizing against the worst case over an uncertainty set

Adverse-selection-aware quoting skewing quotes using order-flow signals

Collar long put + short call that fixes a price band around a holding

Accumulator / decumulator products for staged accumulation or liquidation

Coupon product a yield-enhancement structured note

Structured product a packaged derivative payoff sold to clients or corporates

- ▶ Gueant (2016) [chosen extension]; El Aoud-Abergel (2015); +1 paper of your choice

- ▶ Gueant, O. (2016). The Financial Mathematics of Market Liquidity. Chapman & Hall/CRC.
- ▶ El Aoud, S., & Abergel, F. (2015). A stochastic control approach to option market making. Market Microstructure and Liquidity, 1(1), 1550006.
- ▶ Cartea, Jaikumar, & Penalva (2015). Algorithmic and High-Frequency Trading.
- ▶ Plus one paper of your choice relevant to the chosen extension.